Protocol-Indexed Observation Laws, Interference Influence, and Dual Observable Quotients

An Operational Bridge from Resolution Geometry to Measurement Experiments

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Abstract

We extend the static observational quotient of Resolution Geometry to a protocol-indexed observation system

$$\mathfrak{F} : \mathcal{P} \times X_{\mathrm{lat}} \longrightarrow X_{\mathrm{obs}}, \qquad \mathfrak{F}_P(x) = \mathfrak{F}(P, x).$$

For each protocol P, the latent quotient $Q_P = X_{\mathrm{lat}}/\sim_P$ classifies realizations that the protocol cannot distinguish. Dually, for each declared preparation x, the protocol quotient $\mathcal{I}_x = \mathcal{P}/\equiv_x$ classifies interventions that have the same observable effect. Both quotients satisfy exact universal factorization properties. This separates three operations often conflated under the word measurement: intervention changes the generated observable law; readout and coarse-graining transform an already generated law; and record-dependent continuation is a separate dynamical kernel.

For a two-path observable law we introduce the complex interference coefficient Γ_P by

$$I_P(\phi) = I_{1,P} + I_{2,P} + 2\,\mathrm{Re}(\Gamma_P e^{i\phi}).$$

Changes in $|\Gamma_P|$ and $\arg \Gamma_P$ give operational coordinates for visibility and phase influence without specifying a photon ontology or deriving quantum mechanics. A matched-marginal attack proves a sharp identifiability limit: direct attenuation, Gaussian phase mixing, and a symmetric two-point phase mixture can generate exactly the same unconditional phase-scan law. Consequently, visibility reduction alone does not identify back-action or a unique microscopic mechanism. Repeated conditional statistics, recorded marker channels, and higher residual moments separate the tested mechanism classes. The final deterministic attack used seed 20260819, sampled 240000 configurations, and passed all 15 registered checks. The resulting principle is: quotienting describes distinguishability, protocol variation describes observable influence, and a record-dependent transition kernel describes back-action.

Contents

1 Aim and scope

Experimental outcomes can depend on how an experiment is performed. The purpose of this paper is to represent that dependence without identifying it with conscious observation, passive information loss, or a unique microscopic mechanism. The central object is not one observation map but a family of observable laws indexed by declared interventions.

The construction is deliberately operational. It supplies an exact bridge to interference experiments because phase, intensity, visibility, distinguishability, and conditioned records can be calculated. It does not derive the Hilbert-space formalism, the Born rule, unitary dynamics, collapse, or a latent ontology of light. Quantum mechanics may provide a physical realization of the observable system, but it is not assumed to be the source of the quotient and factorization statements proved here.

2 Protocol-indexed observation systems

Definition 2.1 (Interventional observation system)**.** An interventional observation system consists of a protocol space $\mathcal{P}$, a latent realization space X_{lat}, an observable-law space X_{obs}, and a map

$$\mathfrak{F} : \mathcal{P} \times X_{\mathrm{lat}} \longrightarrow X_{\mathrm{obs}}.$$

For fixed $P \in \mathcal{P}$, write $\mathfrak{F}_P(x) = \mathfrak{F}(P, x)$.

A complete experimental description may be decomposed as

$$P = (s, u, m, d, c),$$

where s denotes declared preparation variables, u an intervention, m a marker or coupling configuration, d a detector/readout map, and c a conditioning rule. This decomposition is not assumed unique; its purpose is to keep causal roles distinct.

Definition 2.2 (Latent equivalence under a protocol)**.** For fixed P, define

$$x \sim_P x' \quad \Longleftrightarrow \quad \mathfrak{F}_P(x) = \mathfrak{F}_P(x').$$

The corresponding quotient and projection are

$$Q_P = X_{\mathrm{lat}}/\sim_P, \qquad \pi_P : X_{\mathrm{lat}} \to Q_P.$$

Theorem 2.3 (Protocol-wise observable factorization)**.** *Let $A_P : X_{\mathrm{lat}} \to Y$ be constant on the fibres of $\mathfrak{F}_P$. Then there is a unique map $\bar{A}_P : Q_P \to Y$ satisfying*

$$A_P = \bar{A}_P \circ \pi_P.$$

Conversely, every map of this form is invariant under $\sim_P$.

Proof. Set $\bar{A}_P([x]_P) = A_P(x)$. Fibre constancy gives well-definedness; the identity follows pointwise; uniqueness follows from surjectivity of π_P. The converse is immediate. □

Definition 2.4 (Protocol-effect equivalence)**.** For a fixed declared realization x, define

$$P \equiv_x P' \quad \Longleftrightarrow \quad \mathfrak{F}_P(x) = \mathfrak{F}_{P'}(x).$$

The protocol-effect quotient is

$$\mathcal{I}_x = \mathcal{P}/\equiv_x, \qquad \kappa_x : \mathcal{P} \to \mathcal{I}_x.$$

Theorem 2.5 (Protocol-effect factorization)**.** *Let $B_x : \mathcal{P} \to Z$ be constant whenever two protocols have the same observable effect on x. Then there is a unique map $\bar{B}_x : \mathcal{I}_x \to Z$ such that*

$$B_x = \bar{B}_x \circ \kappa_x.$$

Proof. This is the same quotient universal property applied in the protocol variable: $\bar{B}_x([P]_x) = B_x(P)$ is well defined and unique. □

Remark 2.6 (Two complementary quotients)*.* Q_P asks which latent realizations are indistinguishable under a fixed protocol. The dual quotient $\mathcal{I}_x$ asks which protocols are indistinguishable by their effect on a fixed preparation. The first formalizes resolution; the second formalizes observable influence.

3 Declared preparation and interventional comparison

Saying that two protocols act on the "same state" is stronger than an experiment normally warrants. We therefore introduce a preparation projection

$$s : X_{\mathrm{lat}} \to X_{\mathrm{prep}}.$$

A declared preparation s_0 identifies the fibre $s^{-1}(s_0)$. Protocols are compared through conditional observable laws

$$\mathfrak{F}_P(\,\cdot \mid s = s_0).$$

This keeps controlled preparation variables fixed without asserting identity of all latent microscopic variables across mutually exclusive interventions.

Definition 3.1 (Observable protocol influence)**.** Let d_{obs} be a declared separating discrepancy on the relevant class of observable laws. The influence discrepancy is

$$\delta(P, P'; s_0) = d_{\mathrm{obs}}(\mathfrak{F}_P(\,\cdot \mid s_0), \mathfrak{F}_{P'}(\,\cdot \mid s_0))\,.$$

An intervention is observationally effective on s_0 when $\delta(P, P'; s_0) > 0$.

Remark 3.2 (No untyped subtraction). The notation $\mathfrak{F}_{P'} - \mathfrak{F}_P$ is not used unless X_{obs} has an explicitly chosen linear or affine structure. The quotient relation or a declared discrepancy is valid for abstract process laws.

4 Intervention, readout, conditioning, and continuation

Let $\mathcal{L}_{u,m}$ denote a pre-readout observable process law generated after the intervention and marker configuration. A detector is represented by a pushforward

$$d_* \mathcal{L}_{u,m},$$

and a conditioning rule produces

$$d_* \mathcal{L}_{u,m}(\,\cdot \mid c).$$

These operations have different meanings.

Proposition 4.1 (Post-readout maps do not cause pre-readout differences)**.** *Let d be a fixed readout map. If $\mathcal{L} = \mathcal{L}'$, then $d_*\mathcal{L} = d_*\mathcal{L}'$. Hence a common passive readout map cannot recover a mechanism distinction absent from its input law, and it cannot be the cause of a difference already present before readout.*

Proof. Pushforward is a function of the input measure. Equal measures have equal pushforwards. □

A record-dependent continuation is a separate transition kernel, for example

$$T_P(dx_{t+1} \mid x_{\le t}, r_{\le t}).$$

If the record is absent from the conditioning variables, the intervention is open loop. If the record changes the transition law, the system contains feedback or back-action. Postselection alone changes a conditional ensemble; it does not by itself establish a changed transition kernel.

5 Two-path observable coefficient

Definition 5.1 (Complex interference coefficient)**.** A two-path phase-scan law is written

$$I_P(\phi) = I_{1,P} + I_{2,P} + 2\,\mathrm{Re}(\Gamma_P e^{i\phi}),$$

where $\Gamma_P \in \mathbb{C}$ is the observable interference coefficient.

For real amplitudes A, B and $\Gamma_P = AB\eta_P e^{i\theta_P}$,

$$I_P(\phi) = A^2 + B^2 + 2AB\eta_P \cos(\phi + \theta_P).$$

Thus η_P measures retained normalized cross-term magnitude and θ_P its phase. When the path intensities are fixed, the visibility is

$$V_P = \frac{2|\Gamma_P|}{I_{1,P} + I_{2,P}}.$$

Proposition 5.2 (Distinct signatures)**.** *Within the preceding family, changing $|\Gamma_P|$ changes visibility, whereas changing $\arg\Gamma_P$ translates the fringe phase without changing its visibility. Therefore deterministic phase displacement alone cannot reproduce a pure visibility change with fixed path intensities.*

Proof. The maximum and minimum intensities are $I_{1,P} + I_{2,P} \pm 2|\Gamma_P|$. Their normalized difference depends only on $|\Gamma_P|$, while the maximizing phase depends on $\arg\Gamma_P$. $\square$

The usual visibility–distinguishability relation is safely stated as an inequality

$$V^2 + D^2 \leq 1.$$

Equality may hold in ideal saturating models. It is not assumed as a universal consequence of Resolution Geometry.

6 Matched-marginal non-identifiability

The next result is the decisive final attack. It asks whether one visibility curve identifies the mechanism that produced it.

Theorem 6.1 (Matched-marginal mechanism non-identifiability)**.** *Fix $0 < \eta \leq 1$. The following three operational mechanisms have the same unconditional first-order phase-scan law:*

$$I_A(\phi) = A^2 + B^2 + 2AB\eta\cos\phi, \tag{1}$$

$$I_B(\phi) = \mathbb{E}_\alpha[A^2 + B^2 + 2AB\cos(\phi + \alpha)], \quad \alpha \sim N(0, \sigma^2), \tag{2}$$

$$I_C(\phi) = \mathbb{E}_\alpha[A^2 + B^2 + 2AB\cos(\phi + \alpha)], \quad \alpha \in \{-\theta, +\theta\} \tag{3}$$

with equal probabilities, provided

$$\sigma^2 = -2\log\eta, \qquad \theta = \arccos\eta.$$

Consequently, an unconditional visibility measurement cannot distinguish direct cross-term attenuation from either tested unresolved phase-mixture mechanism.

Proof. For the Gaussian model, $\mathbb{E}(e^{i\alpha}) = e^{-\sigma^2/2} = \eta$, hence $\mathbb{E}\cos(\phi + \alpha) = \eta\cos\phi$. For the symmetric two-point model, odd terms cancel and

$$\frac{1}{2}\cos(\phi + \theta) + \frac{1}{2}\cos(\phi - \theta) = \cos\phi\cos\theta = \eta\cos\phi.$$

Substitution yields $I_A = I_B = I_C$. $\square$

Corollary 6.2 (Visibility does not identify back-action)**.** *A visibility reduction alone does not entail a unique microscopic mechanism and does not by itself establish record-dependent back-action.*

Proof. The theorem supplies distinct mechanisms with the same visibility law, including unresolved open-loop phase mixtures that require no record-conditioned continuation. □

7 What additional observations can separate

Matched first marginals need not imply matched richer process laws.

Proposition 7.1 (Repeated conditional statistics)**.** *In the synthetic family of the final attack, direct deterministic attenuation has zero residual variance conditional on* (ϕ, η)*, while unresolved Gaussian and symmetric phase mixtures have positive conditional residual variance away from degenerate parameter values. Their fourth residual moments can also differ. Hence repeated conditional observations can separate the tested mechanism classes even when their unconditional means coincide.*

Proof. For direct attenuation the normalized cross term equals $\eta \cos\phi$ identically. Under a nondegenerate random phase it is $\cos(\phi + \alpha)$, which is not almost surely constant conditional on (ϕ, η). Distinct phase distributions generally induce distinct higher conditional moments. □

Proposition 7.2 (Recorded complementary channels)**.** *Let*

$$I_{\pm}(\phi) = \frac{1}{2}\big(1 + c\cos(\phi \pm \theta)\big).$$

The conditional channels can differ, while forgetting the marker gives

$$I_{+}(\phi) + I_{-}(\phi) = 1 + c\cos\theta\cos\phi.$$

Thus retaining a marker record may separate sublaws that become identical after record erasure or marginalization.

Proof. Apply the sum formula for cosine. The difference is proportional to $\sin\phi\sin\theta$ and is nonzero generically. □

Remark 7.3 (Causal safeguard)*.* Later conditioning partitions records and changes conditional laws. Recombining all conditioned classes returns the original marginal. This does not require a retroactive change of an already stored unconditional distribution.

8 Second-order shadows remain derived

For every protocol, a second-order construction is a derived map

$$S_2(\mathfrak{F}_P) = (\Sigma_{R,P}, \Sigma_{N,P}, C_P).$$

It is not the full observation law. The general quotient theorem gives

$$S_2 \circ \mathfrak{F}_P = \overline{S}_{2,P} \circ \pi_P.$$

The earlier saturation result closes the generic continuous second-order gauge quotient on its regular stratum through covariance spectra and mixed trace invariants. It does not close higher moments, temporal memory, conditioned records, or protocol-dependent continuation. The matched-marginal theorem shows concretely why those layers must remain distinct.

9 Observation architecture theorem

Theorem 9.1 (Interventional observation architecture)**.** *An operational observation system separates into the following structures:*

1. *a protocol-indexed observable law* $\mathfrak{F} : \mathcal{P} \times X_{\text{lat}} \to X_{\text{obs}}$*;*
2. *for each protocol, a latent distinguishability quotient* $Q_P = X_{\text{lat}}/{\sim_P}$*;*
3. *for each declared preparation, a protocol-effect quotient* $\mathcal{I}_x = \mathcal{P}/{\equiv_x}$*;*
4. *derived observable shadows* $D(\mathfrak{F}_P)$*, each factoring through* Q_P*;*
5. *optional record-dependent continuation kernels* $T_P(dx_{t+1} \mid x_{\leq t}, r_{\leq t})$*.*

Quotienting, intervention, readout, conditioning, and back-action are therefore mathematically distinct operations.

Proof. Items 1–3 are the definitions and Theorems 2.3 and 2.5. Item 4 follows by applying Theorem 2.3 to $D \circ \mathfrak{F}_P$. Item 5 is additional dynamical data and is not derivable from a static quotient alone. The domains and codomains of the five operations differ, so they cannot be identified without extra structure. □

Corollary 9.2 (Exact meaning of observation influence)**.** *Observation influence is operationally represented by protocol dependence of the observable process law. Back-action is the narrower property that an actual record enters a subsequent transition kernel. Passive coarse-graining is neither of these: it is a transformation of an already generated law.*

10 Final attack battery

The analytic matched-marginal theorem does not depend on simulation. A deterministic Python battery tested implementation consistency, finite-sample recovery, protocol quotienting, causal separation, and latent-extension invariance. The final run used seed 20260819 and 240000 randomized configurations. All 15 registered checks passed.

Attack	Result	Interpretation
Direct vs. Gaussian marginal	pass	Analytic phase-scan laws coincide.
Direct vs. symmetric marginal	pass	Analytic phase-scan laws coincide.
Empirical mean recovery	pass	Binned RMSEs 0.01645 and 0.01597.
First-order identification	fail by design	One marginal cannot select a mechanism.
Conditional residual variance	pass	Variances 0.32464 and 0.32582 separate mixtures from direct attenuation.
Fourth residual moment	pass	Relative gap 0.24923 separates the tested mixtures.
Conditioned marker channels	pass	Mean absolute channel separation 0.99452.
Record forgetting	pass	Complementary channels restore the matched marginal.
Common coarse-graining	pass	Cannot recover an absent mechanism distinction.
Protocol-effect quotient	pass	Five labels form four observable-effect classes.
Causal recombination	pass	Conditioning preserves the recombined marginal.
Silent latent extension	pass	Seven hidden coordinates remain invisible.

Here "fail by design" is the successful discovery of non-identifiability, not a failed registered check. The battery correctly rejects the claim that one visibility curve identifies a unique mechanism.

11 Reproducibility record

The theorem ladder is analytic. Computational artifacts are reproducibility witnesses and boundary attacks, not premises of the proofs.

The prior deterministic families are:

1. fixed-protocol uniqueness battery, seed `20260805`;
2. duration/clock batteries, seed `20260806`;
3. saturation battery for generic orbit closure;
4. observation-law factorization battery, seed `20260818`;
5. light observation-influence battery, seed `20260819`;
6. final matched-marginal protocol-influence attack, seed `20260819`.

The saturation and observation-law artifacts are:

- `latent_observation_saturation.py`;
- `latent_observation_saturation_demo.json`;
- `latent_observation_saturation_battery.json`.

The observation-law battery records class count; rank–nullity; projector idempotence and symmetry; law factorization; fibre annihilation; hidden-extension invariance of class count and quotient law; reparameterization covariance; rejection of strict unobservable refinement; detection of information-losing coarsening; factorization of random observables; rejection of a generic latent function; contrast-rank bound and reconstruction; and a witness that a tolerance relation need not be transitive. It passed all 16 checks, with observable rank 4, null rank 3, and contrast rank 3. Its recorded quotient transition law is

$$\begin{pmatrix} 0.7 & 0.2 & 0.1 & 0 \\ 0.1 & 0.7 & 0.1 & 0.1 \\ 0.25 & 0.25 & 0.25 & 0.25 \\ 0 & 0.2 & 0.3 & 0.5 \end{pmatrix}.$$

The final attack artifacts are:

- `final_protocol_influence_attack.py`;
- `final_protocol_influence_attack.json`.

They contain the complete seed, sample count, thresholds, check outcomes, summary metrics, protocol-effect classes, and verdict. All numerical statements in this section are metadata; no analytic theorem depends on a random sample.

12 Non-claims

Non-claim 12.1. This paper does not derive quantum mechanics, the Born rule, Hilbert-space kinematics, wavefunction collapse, or a microscopic ontology of light. It does not identify observation with consciousness. It does not infer physical back-action from visibility reduction alone. It does not claim that every protocol change is a physical disturbance, nor that one marginal statistic identifies a unique mechanism.

Non-claim 12.2. The protocol quotient classifies observable effects relative to a declared preparation and observation family. Enlarging that family may refine the quotient. Silent latent extensions remain underdetermined, and the framework does not select a preferred latent dimension or topology.

Non-claim 12.3. The statement that observable identification precedes latent reconstruction is a logical ordering of identifiable structures. It is not, by itself, a theorem that measurement is ontologically more fundamental than spacetime.

13 Conclusion

The static observation quotient answers what a fixed protocol can distinguish. It does not by itself express how changing an experiment changes its observable law. The required extension is a protocol-indexed family $\mathfrak{F}_P$, together with two universal quotients: Q_P in the latent variable and $\mathcal{I}_x$ in the protocol variable. In two-path experiments, the complex coefficient Γ_P gives an explicit operational coordinate for changes in cross-term magnitude and phase.

The final attack closes an important overclaim. Direct attenuation and distinct unresolved phase mixtures may have exactly the same visibility curve. Therefore observation influence is identifiable only relative to a sufficiently rich family of records and interventions. Repeated conditional statistics, marker channels, higher moments, and temporal tests can refine the protocol quotient; a single marginal cannot.

Quotienting describes distinguishability.
Protocol variation describes observable influence.
Readout and conditioning transform or partition observable laws.
Record-dependent transition kernels describe dynamical back-action.
None of these statements reconstructs a unique latent ontology.

A Experimental Protocol for Discriminating Observable-Influence Mechanisms in a Two-Path Interferometer

A.1 Purpose

The purpose of this experiment is not to determine a microscopic ontology of light.

Instead, the goal is to determine which classes of observable-law mechanisms are distinguishable by progressively enriched measurement protocols.

In particular, we investigate whether a reduction in interference visibility uniquely identifies a physical back-action mechanism, or whether multiple operationally distinct protocols generate identical unconditioned observable laws.

The experiment is designed as a protocol-discrimination study within the framework of protocol-indexed observation laws.

A.2 Fixed Apparatus

All measurements are performed using a single hardware configuration.

1. Single-photon source or strongly attenuated coherent source.
2. Two-path interferometer (double slit or Mach–Zehnder geometry).
3. Position-sensitive detector: CCD, EMCCD, SPAD array, or equivalent.
4. Programmable path element capable of introducing:
   - deterministic phase shifts,
   - random phase modulation,
   - weak path markers.
5. Optional marker readout channel (polarization analyzer, auxiliary detector, path-correlated ancilla detector, etc.).

No hardware element is added, removed, or repositioned between protocols.
Only protocol variables are modified.

A.3 Observable Coordinate

For each protocol P the measured intensity law is modeled as

$$I_P(\phi) = I_{1,P} + I_{2,P} + 2\,\mathrm{Re}\Big(\Gamma_P e^{i\phi}\Big)\,,$$

where

$$\Gamma_P = |\Gamma_P| e^{i\theta_P}$$

is the experimentally inferred interference coefficient.

The protocol therefore acts on

$$P \longmapsto \Gamma_P.$$

Changes in

$$|\Gamma_P|$$

modify visibility, while changes in

$$\theta_P$$

modify fringe position.

A.4 Protocol Variables

Each protocol is represented as

$$P = (s, u, m, d, c),$$

where

- s: declared preparation settings,
- u: intervention applied within the interferometer,
- m: marker configuration,
- d: detector/readout map,
- c: conditioning rule.

The preparation settings remain fixed throughout the experiment. Only (u, m, d, c) vary.

A.5 Protocol A: Baseline Interference

Intervention

$$u = u_0$$

(no marker, no added phase fluctuation).

Readout

$$d = d_{\text{pos}}$$

(position only).

Conditioning None.

Expected Law

$$I_A(\phi) = A^2 + B^2 + 2AB\cos\phi.$$

The estimated coefficient is

$$\Gamma_A = AB.$$

This protocol provides the reference visibility

$$V_0.$$

A.6 Protocol B: Open-Loop Phase Mixing

Intervention Random phase modulation is applied in one arm.

Two versions may be used:

$$\alpha \sim N(0, \sigma^2)$$

or

$$\alpha = \pm\theta$$

with equal probability.

Readout Position only.

Conditioning None.

Expected Law

$$I_B(\phi) = A^2 + B^2 + 2AB\eta \cos\phi.$$

Visibility is reduced.

The protocol intentionally reproduces the matched-marginal conditions of the non-identifiability theorem.

A.7 Protocol C: Weak Which-Way Marker

Intervention A weak path marker is introduced.

Examples include

- polarization tagging,
- weak ancilla coupling,
- auxiliary path detector.

Readout

$$d = d_{\text{pos+mark}}.$$

Both screen position and marker value are recorded.

Conditioning Modes

1. Unconditioned analysis.
2. Marker-conditioned analysis.

Expected Outcome Unconditioned statistics may exhibit reduced visibility.

Conditioned statistics may exhibit complementary fringes

$$I_+(\phi), \qquad I_-(\phi),$$

satisfying

$$I_+ + I_- = I_{\text{uncond}}.$$

This protocol tests whether additional recorded channels refine the observable law.

A.8 Protocol D: Mechanism-Separation Protocol

Protocol D uses the same hardware and the same target visibility as Protocols B and C.

The distinction lies entirely in the additional recorded information.

For every detection event, record

$$(\phi, x, r),$$

where

- ϕ: phase setting,
- x: detector position,
- r: marker or auxiliary channel.

The following statistics are then computed.

A.8.1 Conditional Residual Variance

Define

$$R = Y - \mathbb{E}[Y \mid \phi, \eta],$$

where Y is the normalized interference contribution.

Estimate

$$\mathrm{Var}(R \mid \phi, \eta)\,.$$

Predictions:

$$\text{direct attenuation} \Rightarrow 0, \tag{4}$$

$$\text{phase mixture} \Rightarrow > 0. \tag{5}$$

A.8.2 Higher Conditional Moments

Estimate

$$\mathbb{E}\left[R^4 \mid \phi, \eta\right].$$

Different hidden phase distributions may possess identical means and variances while exhibiting different fourth moments.

A.8.3 Conditional Channel Structure

Estimate

$$I(x \mid r = +), \qquad I(x \mid r = -).$$

Determine whether

$$I(x) = \sum_r P(r) I(x \mid r)$$

is composed of distinguishable conditioned sublaws.

A.9 Primary Hypotheses

H1 Visibility reduction alone does not identify a unique mechanism.

H2 Distinct operational mechanisms can produce identical unconditioned interference laws.

H3 Conditional channels, repeated statistics, or higher moments can separate mechanism classes that cannot be separated from a single visibility scan.

H4 Observable influence is represented operationally through protocol dependence of the observable law

$$P \longmapsto \Gamma_P,$$

not merely by post-processing or coarse-graining.

A.10 Expected Outcome

The experiment is expected to demonstrate:

1. identical hardware can generate distinct observable laws through protocol variation alone;
2. identical visibility curves can arise from operationally different mechanisms;
3. richer protocols can separate mechanisms that are observationally indistinguishable under position-only detection;
4. protocol dependence, distinguishability structure, and record-dependent continuation constitute separate experimental layers.

The experiment therefore tests observable-law structure rather than latent ontology.

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